SE444 – Artificial Intelligence

Quiz 4: Bayesian Network Inference

ANSWER KEY

Lecture 11: Bayesian Networks & Probabilistic Inference

Time: 30 minutes Total: 15 Questions

Question 1: Inference Objective [Conceptual]

When performing inference in a Bayesian network to compute $P(\text{Query} \mid \text{Evidence})$, what is the primary computational task?

A. Finding the most likely value of the query variable

B. **Marginalizing over hidden variables while conditioning on evidence** ✓

C. Building a new graphical model

D. Computing the prior probabilities of all nodes

E. Maximizing the joint probability distribution

Correct Answer: B

Question 2: Identifying Hidden Variables [Conceptual]

Consider a Bayesian Network with variables $\{X, Y, Z, W, V\}$. For the query $P(X \mid W = \text{true}, V = \text{false})$, identify the hidden variables.

A. X, W, V

B. Y, Z ✓

C. W, V

D. X, Y, Z

E. No hidden variables exist

Correct Answer: B

Question 3: Enumeration Complexity [Computational]

A Bayesian network has 6 binary hidden variables. Using inference by enumeration, how many joint probability assignments must be evaluated?

A. 6

B. 12

C. 36

D. **64** ✓

E. 128

Correct Answer: D ($2^6 = 64$)

Question 4: Understanding the Enumeration Formula [Conceptual]

In the inference formula:

$$P(Q \mid e) = \frac{\sum_h P(Q, e, h)}{\sum_{q'} \sum_h P(q', e, h)}$$

What role does the denominator play?

A. It computes the prior probability of the query

B. **It serves as a normalization constant to ensure probabilities sum to 1** ✓

C. It eliminates the hidden variables

D. It represents the conditional probability tables

E. It calculates the marginal probability of hidden variables

Correct Answer: B

Question 5: Efficiency of Variable Elimination [Conceptual]

What makes Variable Elimination more efficient than direct enumeration for Bayesian network inference?

A. It uses approximation instead of exact computation

B. **It avoids repeated computations by eliminating variables incrementally** ✓

C. It requires fewer CPT entries

D. It can skip hidden variables

E. It converts the DAG into a tree structure

Correct Answer: B

Question 6: Understanding Factors [Conceptual]

In the context of Variable Elimination, what best describes a "factor"?

A. A directed edge in the Bayesian network

B. **A table that assigns numerical values to variable assignments** ✓

C. The probability of observing evidence

D. A variable to be eliminated

E. The output of the inference algorithm

Correct Answer: B

Question 7: Core Operations in VE [Conceptual]

What are the two essential operations used in Variable Elimination?

A. Union and Intersection

B. **Factor Multiplication and Summation (Marginalization)** ✓

C. Addition and Normalization

D. Conditioning and Sampling

E. Join and Projection

Correct Answer: B

Question 8: Result of Marginalization [Computational]

If we have a factor $g(X, Y, Z)$ and we marginalize out variable Y (compute $\sum_Y g(X, Y, Z)$), which variables remain in the resulting factor?

A. Only X

B. Only Y

C. Only Z

D. X **and** Z ✓

E. X, Y, and Z

Correct Answer: D

Question 9: Variable Elimination on a Chain [Critical Thinking]

Given the chain structure $X \to Y \to Z$, to compute $P(Z \mid X = \text{true})$, which variable must Variable Elimination eliminate?

A. X

B. Y ✓

C. Z

D. All three variables

E. None, direct computation is possible

Correct Answer: B

Question 10: Factor Product Computation [Computational]

Given $P(X)$ with $P(X = T) = 0.4$, and $P(Y \mid X)$ with $P(Y = T \mid X = T) = 0.9$ and $P(Y = T \mid X = F) = 0.3$. Calculate $P(X = T, Y = T)$.

A. 0.12

B. 0.27

C. **0.36** ✓

D. 0.40

E. 0.54

Correct Answer: C ($P(X = T) \times P(Y = T \mid X = T) = 0.4 \times 0.9 = 0.36$)

Question 11: Normalization Calculation [Computational]

Variable Elimination produces $P(B = T, e) = 0.12$ and $P(B = F, e) = 0.28$. What is $P(B = T \mid e)$?

A. 0.12

B. 0.28

C. **0.30** ✓

D. 0.40

E. 0.70

Correct Answer: C ($\frac{0.12}{0.12+0.28} = \frac{0.12}{0.40} = 0.30$)

Question 12: Conditional Independence and Inference [Conceptual]

If variables X and Y are d-separated given Z in a Bayesian network, what inference simplification is possible?

A. $P(X \mid Y, Z) = P(X \mid Z)$ – **knowing Y adds no information about X given Z** ✓

B. $P(X, Y) = P(X)P(Y)$ – unconditional independence

C. $P(Z \mid X, Y) = P(Z)$ – Z is independent of both

D. X and Y must share a common parent

E. The network can be simplified to a tree

Correct Answer: A

Question 13: Importance of Elimination Order [Critical Thinking]

In Variable Elimination, why is the order in which variables are eliminated important?

A. Different orders produce different final answers

B. **The order determines the size of intermediate factors, affecting computational cost** ✓

C. Some orders may cause the algorithm to fail

D. The order must match the topological sort of the DAG

E. It changes the conditional independence relationships

Correct Answer: B

Question 14: Diagnostic Reasoning [Critical Thinking / Computation]

A cybersecurity system has:

- $P(\text{Attack}) = 0.02$ (2% of traffic is malicious)
- $P(\text{Alert} \mid \text{Attack}) = 0.95$ (95% detection rate)
- $P(\text{Alert} \mid \neg\text{Attack}) = 0.08$ (8% false alarm rate)

Given an alert, what is approximately $P(\text{Attack} \mid \text{Alert})$?

A. **About 19%** ✓

B. About 45%

C. About 75%

D. About 95%

E. Cannot be determined

Correct Answer: A
Calculation:

$$P(\text{Attack} \mid \text{Alert}) = \frac{P(\text{Alert} \mid \text{Attack}) \cdot P(\text{Attack})}{P(\text{Alert})}$$

$$\begin{aligned} P(\text{Alert}) &= P(\text{Alert} \mid \text{Attack}) \cdot P(\text{Attack}) + P(\text{Alert} \mid \neg\text{Attack}) \cdot P(\neg\text{Attack}) \\ &= 0.95 \times 0.02 + 0.08 \times 0.98 \\ &= 0.019 + 0.0784 = 0.0974 \end{aligned}$$

$$P(\text{Attack} \mid \text{Alert}) = \frac{0.019}{0.0974} \approx 0.195 \approx 19\%$$

Question 15: Variable Elimination Properties [Multi-Select]

Which of the following statements about Variable Elimination are TRUE? **(Select all that apply)**

A. **VE always produces exact inference results** ✓

B. VE runs in polynomial time on all Bayesian networks

C. **VE processes and removes one hidden variable at a time** ✓

D. VE only works on tree-structured networks

E. **The choice of elimination order can significantly impact performance** ✓

Correct Answers: A, C, E

ANSWER SUMMARY

Q#	Answer	Q#	Answer	Q#	Answer
1	B	6	B	11	C
2	B	7	B	12	A
3	D	8	D	13	B
4	B	9	B	14	A
5	B	10	C	15	A, C, E

— End of Answer Key —